Astro HW 10

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NOTE: Work on the page itself is kinda sparse, but all the code used for the exact results is at the end[1]

10.1 Supernova! Wahoo! Yippee!

a) Time since SN

Given ...

$$\theta = \frac{D}{d}$$

$$\theta = 1.2\text{arcmin}, \qquad d = 8.5\text{kpc}, \qquad v_{snr} = 14000\text{km/s}$$

We find ...

$$D_{snr} = (\theta_{snr} \times d_{snr}), \qquad R_{snr} = \frac{D_{snr}}{2}$$

$$t_0 = R_{snr}/V_{snr}$$

$$\boxed{t_0 \approx 3.269 \times 10^9\text{s} \approx 1.03 \times 10^2\text{yr}}$$

b) Neutrinos per sec

Given...

$$F = \frac{L}{4\pi d^2}$$

$$L = 10^{57}\text{neutrinos}, \qquad \Delta T \sim 15\text{s}$$

We find that

$$F = 1.156 \times 10^{15}\text{m}^2$$

$$F_t = \frac{F}{\Delta T}$$

$$\boxed{F_t \approx 7.712 \times 10^{13} \text{ Neutrinos/m }^2\text{s}}$$

[1] pls dont dock me for not showing enough work ;-;

c) Magnitudes and extinction

Given ...

$$M_{bol} = M_{bol\odot} - 2.5\log(L_{bol}/L_\odot), \qquad M_{bol\odot} = 4.74 \qquad L_{bol} = 10^9 L_\odot$$

$$m = M + 5\log(d) - 5, \qquad d = 8.5\text{kpc}$$

$$m_{obsc} = M + 5\log(d) - 5 + A_v \qquad A_v = 30$$

We find ...

$$M_{bol} = 4.474 - 2.5\log(10^9 L_\odot) \approx -17.76$$

$$m_{bol} = -18.1 + 5\log(8500\text{pc}) - 5 \approx -3.113$$

$$m_{ObscBol} = -18.1 + 5\log(8500\text{pc}) - 5 + 30 \approx 26.886$$

$$\boxed{M_{bol} \approx -17.76, \qquad m_{ObscBol} \approx 26.887}$$

This means it is completely impossible to resolve with the naked eye, and, while resolvable with some scopes, still quite difficult.

10.2 Stellar remnants and angular momentum

Given ...

$$I = \frac{2}{5}MR^2$$

$$L = I\omega$$

$$\omega = \frac{\theta}{t}$$

$$P_{sun} = 25.4\text{days}, \qquad r_{wd} = 6000\text{km}$$

We find ...

$$I_{sun} = 2/5 m_\odot r_\odot^2$$

$$\omega_{sun} = (2\pi)/P_{sun}$$

$$L_{sun} = I_{sun}\omega_{sun} \approx 1.1 \times 10^{42}\text{m}^2\text{kg/s}$$

$$I_{wdSun} = 2/5 m_\odot r_{wd}^2$$

By conservation of angular momentum...

$$\omega_{wdSun} = L/I_{wdSun}$$

$$P_{wdSun} = (2\pi/\omega_{wdSun})$$

$$\boxed{P_{wdSun} \approx 163.2\text{s}}$$

10.3 The nature of pulsars

a) Orbiting WDs

Given...

$$P^2 = \frac{4\pi^2}{G(m_1 + m_2)} a^3$$

$$M_{wd} = .7M_\odot, \qquad r_{wd} = 7 \times 10^6\text{m}, \qquad P_{wd} = 1.337\text{s}$$

We find that ...

$$a = \left(\frac{(P_{wd}^2)(2GM_{wd})}{4\pi^2} \right)^{1/3}$$

$$\boxed{a \approx 2 \times 10^6\text{m}}$$

b) Is this possible?

No, this is not possible. This is because $\boxed{a < r_{wd}}$, meaning that the WDs semimajor axis would be literally within each other, and thus they would have to be merged.

c) WD Period

Given ...

$$v_{rot} = \frac{2\pi R}{P}$$

$$R \approx .01R_\odot \left(\frac{M}{.7M_\odot} \right)^{-1/3}$$

We find ...

$$v_{rot} = \frac{2\pi.01R_\odot \left(\frac{M}{.7M_\odot} \right)^{-1/3}}{P}$$

$$v_{rot} = \frac{2\pi \times .01R_\odot}{\text{s}} \left(\frac{P}{1\text{s}} \right)^{-1} \left(\frac{M}{.7M_\odot} \right)^{-1/3}$$

$$\frac{2\pi \times 0.1R_\odot}{c} \approx .145$$

And ...

$$\boxed{.145c \left(\frac{P}{1s} \right)^{-1} \left(\frac{M}{.7M_\odot} \right)^{-1/3} \approx .1c \left(\frac{P}{1s} \right)^{-1} \left(\frac{M}{.7M_\odot} \right)^{-1/3}}$$

d) Breakup Speed

Given ...

$$a_c = \frac{v^2}{r}$$

$$g = \frac{GM}{R^2}$$

We find ...

$$v_{crit} \to \frac{v^2}{R} = \frac{GM}{R^2}$$

$$\frac{v^2}{\cancel{R}} = \frac{GM}{R^{\cancel{2}}}$$

$$\boxed{v = \sqrt{\frac{GM}{R}}}$$

e) Breakup speed from alternative function

Given ...

$$v = \sqrt{\frac{GM}{R}}, \qquad R \approx .01 R_\odot \left(\frac{M}{.7M_\odot}\right)^{-1/3}$$

We find ...

$$v = \sqrt{\frac{GM}{.01R_\odot \left(\frac{M}{.7M_\odot}\right)^{-1/3}}}$$

$$v = \sqrt{\frac{GM}{.01R_\odot}} \sqrt{\left(\frac{M}{.7M_\odot}\right)^{1/3}}$$

$$v = \sqrt{\frac{G}{.01R_\odot}} \sqrt{.7M_\odot} \frac{\sqrt{M}}{\sqrt{.7M\odot}} \left(\frac{M}{.7M_\odot}\right)^{1/6}$$

$$\sqrt{\frac{G}{.01R_\odot}} \sqrt{.7M_\odot}/c \approx 0.0121891c$$

$$\boxed{v_{crit} \approx 0.0121891C \left(\frac{M}{.7M_\odot}\right)^{2/3} \approx .01c \left(\frac{M}{.7M_\odot}\right)^{2/3}}$$

OR Given...

$$v_{crit\mathbf{A}} = \sqrt{\frac{GM_{wd}}{r_{wd}}} \qquad v_{crit\mathbf{B}} \approx .01c \left(\frac{M}{.7M_\odot}\right)^{2/3}$$

$$M_{wd} = .7M_\odot, \qquad r_{wd} = 7 \times 10^6 \text{m}$$

We find that ...

$$\boxed{v_{crit\mathbf{A}} \approx 3.643 \times 10^6 \text{m/s}} \approx \boxed{v_{crit\mathbf{B}} \approx 3.583 \times 10^6 \text{m/s}}$$

Thus, both of the equations return approximately the same answer

f) Is this feasible?

No shot... This is because $\boxed{v_{rot} > v_{crit}}$, meaning that a WD rotating that fast enough to produce the pulsations would break up, as its gravitational acceleration at that speed would be less than that of the centrifugal acceleration.

g) What if it was a NS?

Given ...

$$R = 11\text{km}\left(\frac{M_{ns}}{1.4M_\odot}\right)^{-1/3}$$

$$v_{rot} = \frac{2\pi R}{P}$$

$$v_{crit} = \sqrt{\frac{GM_{NS}}{r_{NS}}}$$

$$M_{ns} = 1.4M_\odot, \qquad r_{NS} = 10\text{km}$$

We find...

$$v_{rot} = \frac{2\pi 11\text{km}\left(\frac{M_{ns}}{1.4M_\odot}\right)^{-1/3}}{P} \boxed{\approx 51694.1\text{m/s}}$$

$$v_{crit} = \sqrt{\frac{GM_{NS}}{11\text{km}\left(\frac{M_{ns}}{1.4M_\odot}\right)^{-1/3}}} \boxed{\approx 1.29964 \times 10^8\text{m/s}}$$

This means that a NS spinning with that rate *does* make sense, this is because $\boxed{v_{rot} < v_{crit}}$, meaning that the NS will *not* break apart, and could have a pulsation period of 1.337s.

10.4 Mass models of the Milky Way Galaxy

Given ...

$$v^2 = \frac{GM}{R} \rightarrow M = \frac{v^2 r}{G}$$

We find ...

$$M_0 = \frac{v_0^2 r_0}{G}, \qquad M_1 = \frac{v_1^2 r_1}{G}, \qquad M_2 = \frac{v_2^2 r_2}{G}, \qquad M_3 = \frac{v_3^2 r_3}{G}$$

$v_c(\text{km}^{-1})$	r(kpc)	$M(<r)(kg)$	$M(<r)M_\odot$
255	0.4	1.20×10^{40}	6.048×10^9
192	2.8	4.77×10^{40}	2.400×10^{10}
220	8	1.79×10^{41}	9.003×10^{10}
235	17	4.34×10^{41}	2.183×10^{11}

10.5 Galactic center black hole and S0-2

a)

Given...

$$M_{bh} = \frac{4\pi^2 a^3}{Gp^2}$$

$$a = 960\text{AU}, \qquad p = 15.6\text{yr}$$

We find ...

$$\boxed{M_{bh} \approx 7.22914 \times 10^{36}\text{kg}}$$

b)

Given...

$$r = a(1 \pm e)$$

$$v_a = \left(\frac{GM_{bh}}{a}\frac{1-e}{1+e}\right)^{1/2}$$

$$v_p = \left(\frac{GM_{bh}}{a}\frac{1+e}{1-e}\right)^{1/2}$$

$$a = 960\text{AU}, \qquad M_{bh} = 7.22914 \times 10^{36}\text{kg}, \qquad e = .867$$

We find...

$$\boxed{r_a \approx 2.618 \times 10^{14}\text{m}}, \qquad \boxed{r_p \approx 1.91 \times 10^{13}\text{m}}$$

$$\boxed{v_a \approx 4.892 \times 10^{5}\text{m/s} \approx .16\% \text{ speed of light}}$$

$$\boxed{v_pe \approx 6.867 \times 10^{6}\text{m/s} \approx 2.29\% \text{ speed of light}}$$

c)

Given ...

$$d = r\left(\frac{2M}{m}\right)^{1/3}$$

$$m = 17.5M_{\odot}, \qquad r = 7.4R_{\odot}, \qquad M_{bh} = 7.22 \times 10^{36}\text{kg}$$

We find ...

$$\boxed{d_{rl} \approx 3.842 \times 10^{11}\text{m}}$$

$$\frac{r_p}{d} \approx 50$$

This star is currently *not* in danger of becoming a TDe, this is because $\boxed{d_{pe} > d_{rl}}$, meaning it doesn't come within the BHs RL, and thus is not at risk.

10.6 Extra Credit

Given...

$$A = -\frac{R_0}{2}\frac{d\omega}{dR}, R = R_0$$

$$2 = \frac{(-\frac{R_0}{2} * \frac{d\omega}{dR}) - (A - \omega_0)}{(\frac{R_0}{2} + \frac{d\omega}{dr})}$$

$$v^2 = \frac{GM}{r} \rightarrow w_0 = \sqrt{\frac{GM}{r^2}}$$

We find ...

$$2 = \frac{\omega_0}{-R_0\frac{d\omega}{dR} - \omega_0}$$

$$2 = \frac{\sqrt{\frac{GM}{r^2}}}{-R_0\frac{d\omega}{dR} - \sqrt{\frac{GM}{r^2}}}$$

$$2(-R_0\frac{d\omega}{dR} - \sqrt{\frac{GM}{r^2}}) = \sqrt{\frac{GM}{r^2}}$$

$$\frac{3\sqrt{GM}}{r} = -2R_0\frac{d\omega}{dR}$$

$$\frac{d\omega}{dR}\omega = \frac{\sqrt{GM}}{R^{3/2}}\frac{d\omega}{dr}$$

$$\frac{d\omega}{dR} = \frac{3}{2}R_0^{-5/3}\sqrt{GM}$$

$$\frac{3\sqrt{GM}}{R_0^{2/3}} = -2\frac{3}{2}\,(R_0)\,(\sqrt{GM})(R_0^{-5/3})$$

$$\frac{\cancel{3}\cancel{\sqrt{GM}}}{\cancel{R_0^{2/3}}} = -\cancel{2}\frac{\cancel{3}}{\cancel{2}}\,(\cancel{R_0})\,(\cancel{\sqrt{GM}})(\cancel{R_0^{-5/3}})$$

$$1 = 1$$

Thus, this equation must be true.